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Sheaf cohomology

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Introduction

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Chapter 1

Sheaf theory

1.1 Introduction

Write motivations for sheafs and presheafs

1.2 Presheaves and sheaves

1.2.1 Presheaves

Definition 1.2.1 (Presheaf)**.** Let X be a topological space and $\mathsf{Op}(X)$ be its associated posetal category of open sets. For an arbitray category C, a C**-valued presheaf** over X is a functor

$$\mathcal{F} : \mathsf{Op}(X)^{\mathrm{op}} \to \mathsf{C}$$

Less abstractly, $\mathcal{F}$ consists of:

1. For every open set $U \subseteq X$ an object $\mathcal{F}(U)$ in C, named *sections* of $\mathcal{F}$ over U.
2. For every inclussion of open sets $V \subseteq U$ of X, a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in C named *restriction*, satisfying $\mathrm{res}_{U,U}) = \mathrm{id}_U$ and for every $W \subseteq V \subseteq U$ open subsets $\mathrm{res}_{V,W} \circ \mathrm{res}_{U,V} = \mathrm{res}_{U,V}$.

   If the source object is clear, we will write $s|_V$ for $\mathrm{res}_{U,V}(s)$.

⌟

Example 1.2.1 (Constant presheaf)**.** For an arbitrary set S, the **constant presheaf** (of sets) associated to S is the Presheaf defined as $\underline{S}_{\mathrm{pre}}(U) = S$ for all non-empty $U \in \mathsf{Op}(X)$ and $\underline{S}_{\mathrm{pre}}(\varnothing) = \{*\}$ an arbitrary one element set.

Note that this verifies the presheaf conditions trivially since for every open set $U \in \mathsf{Op}(X)$, $\underline{S}_{\mathrm{pre}}(U)$ is a set (namely S) and, if $V \subseteq U$ then we can pick id_S as $\mathrm{res}_{U,V}$.

If we work over a more general category C we require that $\underline{S}_{\mathrm{pre}}(\varnothing)$ is the final object in C, provided it exists. ◊

1.2.2 Sheaves

Definition 1.2.2 (Sheaf)**.** Let X be a topological space. A Presheaf of sets $\mathscr{F}$ is a **sheaf** if for every open set $U \subseteq X$ and open cover $\{U_i\}_{i \in \mathcal{I}}$ of U the following axioms hold:

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections ($\forall i, j \in \mathcal{I}$ $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$) there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I}$ $s|_{U_i} = s_i$.

⌟

Notation. To differentiate between a Presheaf and a Sheaf, we will denote the former using $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the later using $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

We will now see an example of a Presheaf that is not a Sheaf.

Example 1.2.2 (The Constant presheaf is not a Sheaf)**.** Consider a set $X = \{\triangle, \square\}$ endowed with the discrete topology and let $\underline{S}_{\text{pre}}$ be the Constant presheaf over X of an arbitrary set S with more than one element, say $S = \{0, 1\}$ with $0 \neq 1$.

Consider the open cover $\{X_1 = \{\triangle\}, X_2 = \{\square\}\}$ of X and define $0 = s_1 \in \underline{S}_{\text{pre}}(X_1) = S$ and $1 = s_2 \in \underline{S}_{\text{pre}}(X_2) = S$. Since $X_1 \cap X_2 = \varnothing$, $s_1|_{X_1\cap X_2} = s_2|_{X_1\cap X_2}$.

Now, let $s \in \underline{S}_{\text{pre}}(X)$ such that $s|_{U_1} = s_1$ and $s|_{U_2} = s_2$. Since the restrictions from $\underline{S}_{\text{pre}}(X)$ are the identity maps we have that

$$0 = s_1 = s|_{U_1} = s = s|_{U_2} = s_2 = 1$$

which is a contradiction because we assumed $0 \neq 1$.

Therefore, since SH.2 fails, $\underline{S}_{\text{pre}}$ is not a sheaf. ◊

However, we can make it a sheaf by considering *locally constant* applications instead.

Example 1.2.3 (Constant sheaf)**.** Let X be a topological space and A be an object of a category C. For every open set $U \subseteq X$, define the **constant sheaf** over X of A as

What type of category?

$$\underline{S}(U) = \{f : U \to A \mid f \text{ locally constant}\}$$

Note that this is a Presheaf.

To verify that this is indeed a sheaf, let $U \subseteq X$ be a non-empty open set an $\{U_i\}_{i\in\mathcal{I}}$ be an open cover of U (in the case where $U = \varnothing$ the axioms hold trivially).

SH.1 Suppose s and t are two sections of $\underline{S}(U)$ such that $\forall i \in \mathcal{I}$ $s|_{U_i} = t|_{U_i}$.

Note that for every $x \in U$ there exists U_i such that $x \in U_i$.

Therefore $s|_{U_i} = \{x\} = t_{U_i}$ and, since this holds for every $x \in U$, $s = t$.

SH.2 Suppose $\{s_i \mid s_i \in \underline{S}(U_i)\}_{i\in\mathcal{I}}$ is a collection of sections that agree on pairwise intersection.

Define $s : U \to S$ as follows: for each $x \in U$ choose $i \in \mathcal{I}$ such that $x \in U_i$ and set $s(x) = s_i(x)$. Note that s is well-defined since if $x \in U_i \cap U_j$ by hypothesis $s_i(x) = s_j(x)$. We only need to show that $s \in \underline{S}$.

For every $x \in U$ choose U_i such that $x \in U_i$. Since s_i is locally constant there exists an open neighbourhood $V \subseteq U_i$ of x such that s is constant in V. Then since $s|_V = s_i|_V$ we have that s is constant in V, meaning that s is locally constant in U.

◊

We can also construct Sheaves from other data:

Example 1.2.4. Let X be a topological space and let $\{A_p\}_{p\in X}$ be a family of sets.

Given an open set $U \subseteq X$ consider $\prod(U) = \prod_{p\in U} A_p$ and if $V \subseteq U$ define $\mathrm{res}_{U,V}(a_p)_{p\in U} = (a_p)_{p\in V}$, making $\prod$ into a Presheaf.

To check that it is indeed a Sheaf, consider an open covering $\{U_i\}_{i\in\mathcal{I}}$ of U. Let $\{s_i \in \prod(U_i)\}_{i\in\mathcal{I}}$ be such that for every $i,j \in \mathcal{I}$ $\quad s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$. If we write $s_i = (a_{i,p})_{p\in U_i}$ then the compatibility condition means that $a_{i,p} = a_{j,p}$ for every $p \in U_i \cap U_j$. Therefore there exists $s = (a_p)_{p\in U} \in \prod(U)$ such that for every $p \in U_i$ $\quad a_p = a_{i,p}$. ◊

Definition 1.2.3 (Pushforward Sheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X and Y. For any Sheaf $\mathscr{F}$ over X, define the **pushforward sheaf** $f_*\mathscr{F}$ in Y as, for every open subset $V \subseteq Y$, $f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(V))$.

This defines a *pushforward functor* $f_* : \mathsf{Sh}(X) \to \mathsf{Sh}(Y)$. ⌟

Example 1.2.5 (Skyscrapper sheaf)**.** Let X be a topological space with $p \in X$. Consider the inclussion $\iota_p : \{p\} \hookrightarrow X$. Then the **skyscrapper sheaf** $\iota_{p,*}\underline{S}$ is the Pushforward Sheaf of the Constant sheaf $\underline{S}$:

$$\iota_{p,*}\underline{S}(U) = \begin{cases} S & p \in U \\ \{*\} & p \notin U \end{cases}$$

◊

We can extend Definition 1.2.2 to more general categories using the following

Motivation for this definition. Why is it equivalent?

Definition 1.2.4 (Sheaf)**.** Let X be a topological space and let C be a category with products. A C-valued Presheaf $\mathscr{F}$ is a **sheaf** if for every open set U and open covering $\{U_i\}_{i\in\mathcal{I}}$ of U the diagram

$$\mathscr{F}(U) \to \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j) \tag{1.2.1}$$

is an equalizer. ⌟

For sufficienly nice categories we have the following [Stacks, Tag 0073],

Proposition 1.2.1. *Let* C *be a category and* $F : \mathsf{C} \to \mathsf{Set}$ *be a functor verifying:*

1. *F is faithful.*
2. *C has limits and F commutes with them.*
3. *F reflects isomorphisms.*

Then a C-valued Presheaf $\mathscr{F}$ is a Sheaf if and only if the underlying Presheaf of sets $F(\mathscr{F})$ is a Sheaf.

Proof.

$\Longrightarrow$ Suppose $\mathscr{F}$ is a C-valued Sheaf.

Then, for every open set U, $\mathscr{F}(U)$ is the equalizer of the diagram (1.2.1). Therefore since F commutes with limits $F(\mathscr{F}(U))$ is the equalizer of the corresponding diagram in Set.

$\Longleftarrow$ Suppose now that $F(\mathscr{F})$ is a Sheaf.

Let $E \in \mathsf{C}$ be the equalizer of the diagram (1.2.1). Since $\mathscr{F}(U)$ is a Presheaf there exists a morphism $\mathscr{F}(U) \to E$, therefore by applying F we obtain a morphism of sets $F(\mathscr{F}(U)) \to F(E)$.

On the one hand since F commutes with limits $F(E)$ is the equalizer of the corresponding diagram in Set. On the other since $F(\mathscr{F})$ is a Sheaf, it is the equalizer of the same diagram. Therefore the morphism $F(\mathscr{F}(U)) \to F(E)$ is an isomorphism.

Finally since F reflects isomorphisms we have that $\mathscr{F}(U) \to E$ is an isomorphism.

□

In particular Proposition 1.2.1 applies to Sheaves of groups, rings, algebras/modules/vector spaces over a fixed ring, etc.

Since we will be mainly interested in Sheaves of abelian groups and rings, we can restrict ourselves to the study of Sheaves of sets and apply Proposition 1.2.1 when needed.

1.3 Morphisms of (pre)sheaves

Definition 1.3.1 (Morphism of Presheaves)**.** Let $\mathcal{F}, \mathcal{G}$ be two C-valued Presheaves over a topological space X. A **presheaf morphism** $\phi : \mathcal{F} \to \mathcal{G}$ is a natural transformation between the functors $\mathcal{F}$ and $\mathcal{G}$.

Less abstractly, a presheaf morphism $\phi : \mathcal{F} \to \mathcal{G}$ is the a collection of maps $\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Op}(X)}$ such that if $V \subseteq U$ then the following diagram commutes

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow{\phi(V)} & \mathcal{G}(V) \end{array}$$

⌟

Definition 1.3.2 (Morphism of Sheaves)**.** Let $\mathscr{F}, \mathscr{G}$ be two C-valued Sheaves over a topological space X. A **sheaf morphism** $\phi : \mathscr{F} \to \mathscr{G}$ is a Morphism of Presheaves between $\mathscr{F}$ and $\mathscr{G}$. ⌟

With this choice of morphisms, Presheaves and Sheaves valued in a category C over a topological space X form a category.

Notation. We will denote the category of C-valued Presheaves over a topological space X as $\mathsf{PSh}_{\mathsf{C}}(X)$ and the category of C-valued Sheaves over X as $\mathsf{Sh}_{\mathsf{C}}(X)$.

Remark 1.3.1. The category $\mathsf{Sh}_{\mathsf{C}}(X)$ is a full-subcategory of $\mathsf{PSh}_{\mathsf{C}}(X)$ by the definition of Morphism of Sheaves, i.e: the inclusion functor $\iota : \mathsf{Sh}_{\mathsf{C}}(X) \hookrightarrow \mathsf{PSh}_{\mathsf{C}}(X)$ is fully faithful. This means that if $\iota(\alpha)$ is a diagram limiting in $\mathsf{PSh}_{\mathsf{C}}(X)$ then α limits in $\mathsf{Sh}_{\mathsf{C}}(X)$. ▽

Definition 1.3.3 ($\mathcal{H}om$ Sheaf)**.** Let $\mathscr{F}, \mathscr{G} \in \mathsf{Sh}_{\mathsf{Set}}(X)$ be two Sheaves of sets over X. We define the $\mathcal{H}om$ Sheaf (of sets) between $\mathscr{F}$ and $\mathscr{G}$ as

$$\forall U \in \mathsf{Op}(X) \quad \mathcal{H}om(\mathscr{F}, \mathscr{G})(U) = \mathrm{Mor}(\mathscr{F}|_U, \mathscr{G}|_U)$$

⌟

Remark 1.3.2. Let's check that the $\mathcal{H}om$ Sheaf $\mathcal{H}om(\mathscr{F}, \mathscr{G})$ is indeed a Sheaf of sets for any $\mathscr{F}, \mathscr{G} \in \mathsf{Sh}_{\mathsf{Set}}(X)$.

First, we need to verify that it is a Presheaf.

We only need to define the restriction maps and check commutativity. For a fixed open set U, the restriction of a section $s \in \mathcal{H}om(\mathscr{F},\mathscr{G})(U)$ to an open subset $V \subseteq U$ is $s|_V \in \mathrm{Mor}(\mathscr{F}|_V,\mathscr{G}|_V)$ defined as $s|_V(W) = s(W) : \mathscr{F}(W) \to \mathscr{G}(W)$ for every open set $W \subseteq V$. Note that, since W is an open subset of U, commutativity in V is inherited from commutativity in U.

Now, we will check the Sheaf axioms. To do this, consider an open covering $\{U_i\}_{i\in\mathcal{I}}$ of an open set U.

SH.1 Let $\phi,\psi \in \mathcal{H}om(\mathscr{F},\mathscr{G})(U)$ be two sections such that for every $i \in \mathcal{I}$ $\phi|_{U_i} = \psi|_{U_i}$. We want to show that for every open subset $V \subseteq U$ the maps $\phi(V)$ and $\psi(V)$ are the same.

First, note that $\{V \cap U_i\}_{i\in\mathcal{I}}$ is an open cover of V. Then, for any $s \in \mathscr{F}(V)$ consider the restrictions $s|_{V\cap U_i} \in \mathscr{F}(V \cap U_i)$.

Since $\phi|_{U_i} = \psi|_{U_i}$ we have that

$$\phi(V)(s|_{V\cap U_i}) = \phi(V \cap U_i)(s|_{V\cap U_i}) = \psi(V \cap U_i)(s|_{V\cap U_i}) = \psi(V)(s|_{V\cap U_i})$$

Therefore, by applying SH.1 to $\mathscr{G}$ we have that $\phi(V)(s) = \psi(V)(s)$, i.e: $\phi(V) = \psi(V)$.

SH.2 Let $\{\phi_i \mid \phi_i \in \mathcal{H}om(\mathscr{F},\mathscr{G})(U_i)\}$ be a collection of sections that agree on pairwise intersection. We want to show that there exists $\phi \in \mathcal{H}om(\mathscr{F},\mathscr{G})(U)$ such that for every $i \in \mathcal{I}$ $\phi|_{U_i} = \phi_i$.

Let $V \subseteq U$ be an open set and $s \in \mathscr{F}(V)$. For every $i \in \mathcal{I}$ consider $s_i = s|_{V\cap U_i} \in \mathscr{F}(V \cap U_i)$. Applying ϕ_i we get $t_i = \phi_i(V \cap U_i)(s_i) \in \mathscr{G}(V \cap U_i)$.

Since for any $i,j \in \mathcal{I}$ $\phi_i|_{U_i\cap U_j} = \phi_j|_{U_i\cap U_j}$ we have that $t_i|_{V\cap U_i\cap U_j} = t_j|_{V\cap U_i\cap U_j}$. Therefore since the sections $\{t_i\}_{i\in\mathcal{I}}$ agree on pairwise intersection applying SH.2 we have that there exists $t \in \mathscr{G}(V)$ such that $\forall i \in \mathcal{I}$ $t|_{V\cap U_i} = t_i$.

This defines a map $\phi(V) : \mathscr{F}(V) \to \mathscr{G}(V)$ for every open set $V \subseteq U$. We need to show that this map is a Morphism of Sheaves, i.e: that it commutes with restriction maps. Let $W \subseteq V$ be an open set and $s \in \mathscr{F}(V)$. We want to show that $\phi(W)(s|_W) = \phi(V)(s)|_W$.

By construction, $\phi(V)(s)$ is the unique section in $\mathscr{G}(V)$ such that for all $i \in \mathcal{I}$

$$\phi(V)(s)|_{V\cap U_i} = \phi(V \cap U_i)(s|_{V\cap U_i})$$

Then restricting boths sides to $W \cap U_i$

$$(\phi(V \cap U_i)(s|_{V\cap U_i}))\,|_{W\cap U_i} = (\phi(V \cap U_i)(s|_{V\cap U_i}))\,|_{W\cap U_i} = \phi_i(V \cap U_i)(s|_{V\cap U_i})|_{W\cap U_i}$$

Since ϕ_i is a Morphism of Sheaves it commutes with restrictions, giving

$$\phi_i(V \cap U_i)(s|_{V\cap U_i})|_{W\cap U_i} = \phi_i(W \cap U_i)((s|_{V\cap U_i})|_{W\cap U_i}) = \phi_i(W \cap U_i)(s|_{W\cap U_i})$$

Therefore

$$\phi(V)(s)|_{W\cap U_i} = \phi_i(W \cap U_i)(s|_{W\cap U_i})$$

On the other hand, $\phi(W)(s|_W)$ is by definition the unique section in $\mathscr{G}(W)$ such that for all $i \in \mathcal{I}$

$$\phi(W)(s|_W)|_{W\cap U_i} = \phi_i(W \cap U_i)((s|_W)|_{W\cap U_i}) = \phi_i(W \cap U_i)(s|_{W\cap U_i})$$

Therefore we have that for all $i \in \mathcal{I}$

$$\phi(V)(s)|_{W\cap U_i} = \phi(W)(s|_W)|_{W\cap U_i}$$

Applying SH.1 we get $\phi(V)(s)|_W = \phi(W)(s|_W)$.

It only remains to show that $\phi|_{U_i} = \phi_i$. Indeed, for every open subset $V \subseteq U_i$ and $s \in \mathscr{F}(V)$ we have that $\phi(V)(s) = \phi_i(V)(s)$ so applying SH.2 on the open cover $\{V \cap U_j\}_{j\in\mathcal{I}}$ we have that $\phi(V) = \phi_i(V)$.

Note that to check the sheaf axioms we only used that $\mathscr{G}$ is a Sheaf, therefore the hypothesis of $\mathscr{F}$ being a Sheaf may be dropped. ▽

Remark 1.3.3. We can extend $\mathcal{H}om$ Sheaf to other evaluation categories other than Set. However, this will be a sheaf only if the evaluation category is *sufficiently* nice (in particular, if it is complete).

This should be rewritten.

For example, if $\mathscr{F}, \mathscr{G} \in \mathsf{Sh}_{\mathsf{Ab}}(X)$ we can define the $\mathcal{H}om$ sheaf, this time an abelian group Sheaf, by considering the maps $\mathscr{F}|_U \to \mathscr{G}|_U$ as morphisms between abelian groups. To distinguish these $\mathcal{H}om$ sheaves from the Set-valued version, we will denote them by $\mathcal{H}om_{\mathsf{Sh}_{\mathsf{C}}(X)}$, although many times the subscript will be dropped if the category is clear. ▽

Example 1.3.1. If $\mathscr{F}$ is a Sheaf of sets over a topological space X then the $\mathcal{H}om$ Sheaf is $\mathcal{H}om(\underline{\{p\}}, \mathscr{F}) \cong \mathscr{F}$.

First, note that for every open set $U \subseteq X$ $\mathcal{H}om(\underline{\{p\}}, \mathscr{F})(U) = \mathrm{Hom}(\underline{\{p\}}|_U, \mathscr{F}|_U)$. Therefore for every $U \subseteq X$ the elements of the $\mathcal{H}om$ Sheaf at U are morphisms $\phi : \underline{\{p\}}|_U \to \mathscr{F}|_U$, i.e: a collection of maps $\phi(V) : \underline{\{p\}}|_V \to \mathscr{F}|_V$ that commute with the restriction maps for every $V \subseteq U$ open. Furthermore since $\underline{\{p\}}|_V = \{p\}$ the maps $\phi(V)$ are determined by the image of p.

Consider $s^{(V)} = \phi(V)(p) \in \mathscr{F}(V)$. Since ϕ commutes with restriction maps we have that for every $W \subseteq V$ open $\mathrm{res}_{V,W}\, s^{(W)} = s^{(V)}$. Therefore $\{s^{(V)}\}_{V\subseteq U}$ is a compatible family of sections, which means that by SH.2 there exists a unique section $s \in \mathscr{F}(U)$ such that $s|_V = s^{(V)}$ for every $V \subseteq U$ open. In particular taking $V = U$ we have that $s|_U = s$, i.e: the morphism ϕ is uniquely determined by the section $s = \phi(U)(p) \in \mathscr{F}(U)$.

On the other hand, given any section $s \in \mathscr{F}(U)$ we can define a morphism $\phi : \underline{\{p\}}|_U \to \mathscr{F}|_U$ by $\phi(V)(p) = s|_V$ for all $V \subseteq U$ open.

Since this morphism satisfies the compatibility conditions, for every open set $U \subseteq X$ we have a bijection

$$\begin{aligned}\Theta_U : \mathcal{H}om(\underline{\{p\}}, \mathscr{F})(U) &\to \mathscr{F}(U)\\ \phi &\mapsto \phi(U)(p)\end{aligned}$$

It remains to show that Θ_U is a Morphism of Sheaves, i.e: that the following diagram commutes fir every $V \subseteq U$ open.

$$\begin{array}{ccc}\mathcal{H}om(\underline{\{p\}}, \mathscr{F})(U) & \xrightarrow{\Theta_U} & \mathscr{F}(U)\\ \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}}\\ \mathcal{H}om(\underline{\{p\}}, \mathscr{F})(V) & \xrightarrow{\Theta_V} & \mathscr{F}(V)\end{array}$$

Let $\phi \in \mathcal{H}om(\underline{\{p\}}, \mathscr{F})(U)$. Note that since ϕ is a Morphism of Sheaves

$$\begin{aligned}(\Theta_V \circ \mathrm{res}_{U,V})(\phi) &= \Theta_V(\phi|_V) = (\phi|_V)(V)(p) = \phi(V)(p) \\ &= \mathrm{res}_{U,V}\,\phi(U)(p) = (\mathrm{res}_{U,V} \circ \Theta_U)(\phi)\end{aligned}$$

Therefore the collection of maps $\{\Theta_U\}_{U \in \mathsf{Op}(X)}$ defines an isomorphism

$$\mathcal{H}om(\underline{\{p\}}, \mathscr{F}) \cong \mathscr{F}$$

◊

Example 1.3.2. If $\mathscr{F}$ is a Sheaf of abelian groups then $\mathcal{H}om(\underline{\mathbb{Z}}, \mathscr{F}) \cong \mathscr{F}$. ◊

1.4 Stalks and germs.

Definition 1.4.1 (Stalk)**.** Let $\mathcal{F}$ be a Presheaf of sets over a topological space X. The **stalk** of $\mathcal{F}$ at a point $p \in X$ is

$$\mathcal{F}_p = \operatorname*{colim}_{\substack{U \ni p \\ U \text{ open}}} \mathcal{F}(U)$$

where the colimit is over the set of open neighbourhoods U of p.

This defines a **stalkification functor** $(-)_p : \mathsf{PSh}_{\mathsf{Set}}(X) \to \mathsf{Set}$. ⌟

Remark 1.4.1. Since the set of open neighbourhoods of a point is partially ordered by reverse inclussion the Stalk colimit is a directed colimit. Therefore we can describe $\mathcal{F}_p$ as

Explain why this works.

$$\mathcal{F}_p = \{(s, U) \mid p \in U, s \in \mathcal{F}(U)\}/_\sim$$

where $(s, U) \sim (t, V)$ if and only if there exists an open set $W \subseteq U \cap V$ with $p \in W$ such that $s|_W = t|_W$.

For a section $f \in \mathcal{F}(U)$ we call $f_p = [(f, U)] \in \mathcal{F}_p$ the **germ** of f at p. ▽

Example 1.4.1. Following Example 1.2.4 given a Presheaf of sets over a topological space X we can define the Sheaf $\prod(\mathcal{F})$ as $\prod(\mathcal{F})(U) = \prod_{p \in U} \mathcal{F}_p$ for every open set $U \subseteq X$. ◊

Proposition 1.4.1. *Let X be a topological space and let $\mathscr{F}$ be a Sheaf of sets over X. Then for every open set $U \subseteq X$ the map $\mathscr{F}(U) \to \prod_{p \in U} \mathscr{F}_p$ is injective.*

Proof. Let $s, t \in \mathscr{F}(U)$ be two sections that map to the same element in every Stalk $\mathscr{F}_p$ for every $p \in U$. Then by Remark 1.4.1 for every $p \in U$ there exists an open neighbourhood $V^p \subseteq U$ such that $s|_{V^p} = t|_{V^p}$. Therefore since $\{V_p\}_{p \in U}$ is an open covering of U by SH.1 $s = t$. □

Proposition 1.4.2 (Induced Stalk morphism)**.** *Let $\mathcal{F}, \mathcal{G} \in \mathsf{PSh}_{\mathsf{Set}}(X)$ be two Presheaves of sets over X and $\phi : \mathcal{F} \to \mathcal{G}$ a Morphism of Presheaves. Then for every $p \in X$, ϕ induces a morphism of Stalks $\phi_p : \mathcal{F}_p \to \mathcal{G}_p$.*

Proof. We have the following diagram, where the vertical arrows come from the colimit properties.

Therefore, we just need to define for every $[(f, U)] \in \mathcal{F}_p \quad \phi_p([(f, U)]) = [(\phi(U)(f), U)]$. □

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \mathcal{F}_p & \xrightarrow{\phi_p} & \mathcal{G}_p \end{array}$$

To generalize the concept of Stalk to other categories in a similar fashion to Proposition 1.2.1 we have the following result

Proposition 1.4.3. *Let* C *be a category and* $F : \mathsf{C} \to \mathsf{Set}$ *be a functor verifying:*

1. *F is faithful.*
2. *Directed colimits exist in* C *and F commutes with them.*

Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C*-valued Presheaf. Then $\mathcal{F}_p = \operatorname{colim}_{p \in U} \mathcal{F}(U)$ exists in* C *and its underlying set $F(\mathcal{F}_p)$ is equal to the Stalk of sets of $\mathcal{F}$.*

Furthermore the construction $\mathcal{F} \mapsto \mathcal{F}_p$ is functorial.

Proof. Since directed colimits exist in C by Remark 1.4.1 the Stalk $\mathcal{F}_p$ exists in C. Uniqueness of the Stalk follows analogously to Proposition 1.2.1. □

Theorem 1.4.1 (Stalks and isomorphisms)**.** *Let $\mathscr{F}, \mathscr{G}$ be two Sheaves of sets over a topological space X and let $\phi : \mathscr{F} \to \mathscr{G}$ be a Morphism of Sheaves. Then ϕ is an isomorphism if and only if for every $p \in X$ the Induced Stalk morphism $\phi_p : \mathscr{F}_p \to \mathscr{G}_p$ is an isomorphism.*

Proof. Suppose first that ϕ is an isomorphism. Then since taking Stalks is a functor and functors preseve isomorphisms, ϕ_p is an isomorphism.

To prove the converse, suppose that for every $p \in X$ ϕ_p is an isomorphism. We are going to show that for every open set $U \subseteq X$ $\phi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ is an isomorphism by proving that it is an injective and surjective morphism in Set.

For the injectivity, let $s, t \in \mathscr{F}(U)$ be two sections such that $\phi(U)(s) = \phi(U)(t)$. Then for every $p \in U$ we have that $\phi_p(s_p) = \phi_p(t_p)$. Since ϕ_p is an isomorphism in particular is injective, therefore $s_p = t_p$. This means that there exist $V_p \subseteq U$ open neighbourhood of p such that $s|_{V_p} = t|_{V_p}$. Since $\{V_p\}_{p \in U}$ is an open cover of U by SH.1 we have that $s = p$, i.e: $\phi(U)$ is injective.

For surjectivity, consider $t \in \mathscr{G}(U)$. For every $p \in U$ since ϕ_p is surjective we have that there exists $s_p \in \mathscr{F}_p$ such that $\phi_p(s_p) = t_p$. By definition of Stalk we have that there exists an open neighbourhood $U_p \subseteq U$ of p and a section $s^{(p)} \in \mathscr{F}(U_p)$ such that $\phi(U_p)(s^{(p)}) = t|_{U_p}$. Note that on overlaps $U_p \cap U_q$ we have that

$$\phi(U_p \cap U_q)(s^{(p)}|_{U_p \cap U_q}) = t|_{U_p \cap U_q} = \phi(U_p \cap U_q)(s^{(q)}|_{U_p \cap U_q})$$

Therefore since ϕ is injective we have that $s^{(p)}|_{U_p \cap U_q} = s^{(q)}|_{U_p \cap U_q}$. Then by SH.2 there exists a section $s \in \mathscr{F}(U)$ such that $\phi(U)(s) = t$, proving surjectivity. □

Theorem 1.4.1 is Exercise 2.4.D of Vakil. However iso $\neq$ inj + surj in general. Maybe it is better to follow Stalks 007S.

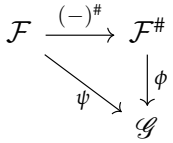
1.5 Sheafification

In Example 1.2.1 we have seen how we can *convert* a Presheaf into a Sheaf. We are now going to see how we can construct Sheaves from Presheaves.

We will first define categorically what **sheafification** ought to be and then we will construct it, therefore showing its existence.

We can restrict ourselves to the case where $\mathsf{C} = \mathsf{Set}$ and use Proposition 1.2.1 when working on more *general* categories.

Definition 1.5.1 (Sheafification)**.** Let $\mathcal{F}$ be a Presheaf of sets over a topological space X. A Morphism of Presheaves $(-)^{\#} : \mathcal{F} \to \mathcal{F}^{\#}$ is the **sheafification** of $\mathcal{F}$ if $\mathcal{F}^{\#}$ is a Sheaf and for any other Sheaf $\mathcal{G}$ and Morphism of Presheaves $\psi : \mathcal{F} \to \mathcal{G}$ there exists a unique Morphism of Sheaves $\phi : \mathcal{F}^{\#} \to \mathcal{G}$ making the following diagram commute



This defines a **sheafification functor** $(-)^{\#} : \mathsf{PSh}_{\mathsf{Set}}(X) \to \mathsf{Sh}_{\mathsf{Set}}(X)$. ⌟

Remark 1.5.1 (Uniqueness of Sheafification). Since Sheafification is defined using an universal property if it exists it is unique (up to unique isomorphism). Therefore we can speak of *the* sheafification of a presheaf instead of *a* sheafification. ▽

Remark 1.5.2. The Sheafification functor is left-adjoint to the inclussion functor $i : \mathsf{Sh}_{\mathsf{Set}}(X) \hookrightarrow \mathsf{PSh}_{\mathsf{Set}}(X)$. This means that for every Presheaf $\mathcal{F}$ and Sheaf $\mathcal{G}$ we have $\mathrm{Hom}_{\mathsf{Sh}}(\mathcal{F}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathsf{PSh}}(\mathcal{F}, i(\mathcal{G}))$. ▽

Example 1.5.1. For any Sheaf $\mathscr{F}$ its Sheafification is itself: $\mathscr{F}^{\#} = \mathscr{F}$ with $(-)^{\#} : \mathscr{F} \to \mathscr{F}^{\#}$ the identity morphism. ◊

Before constructing the Sheafification functor we will take a detour to study the inverse image Sheaf and its relation to Stalks.

Lemma 1.5.1. *Let $f : X \to Y$ be a continuous map between two topological spaces X and Y. Let $\mathcal{G}$ be a Sheaf over Y.*

Then $f^{-1}_{\mathrm{pre}}\mathcal{G}$ defined as for every open set $U \subseteq X \quad f^{-1}_{\mathrm{pre}}\mathcal{G}(U) = \mathrm{colim}_{V \supset f(U)} \mathcal{G}(V)$ is a Presheaf but not a Sheaf.

Definition 1.5.2 (Inverse image Sheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X and Y. Let $\mathcal{G}$ be a Sheaf of sets over Y. Then **inverse image** Presheaf of $\mathcal{G}$ by f is $f^{-1}\mathcal{G} = (f^{-1}_{\mathrm{pre}}\mathcal{G})^{\#}$.

This defines an *inverse image functor* $f^{-1} : \mathsf{Sh}(Y) \to \mathsf{Sh}(X)$. ⌟

Example 1.5.2. Let Y be a topological space and $p \in Y$. Then the Inverse image Sheaf $f^{-1}\mathcal{G}$ corresponds to the Stalk $\mathcal{G}_p$. ◊

Remark 1.5.3. ▽

Proposition 1.5.1. *For every continuous map $f : X \to Y$ the functors induced by the Pushforward Sheaf and the Inverse image Sheaf are adjoint.*

Remark 1.5.4. Proposition 1.5.1 tells us that taking Stalks and the Skyscrapper sheaf are adjoint functors. ▽

We will now construct the Sheafification of a Presheaf $\mathcal{F}$. Our goal is to prove the following.

Theorem 1.5.1 (Construction of the Sheafification)**.** *Let $\mathcal{F}$ be a Presheaf of sets over a topological space X. Then*

$$\mathcal{F}^{\#}(U) = \left\{(f_p \in \mathcal{F}_p)_{p\in U} \mid f_p \text{ verifies } (*)\right\} \tag{1.5.1}$$

where $(*)$ is the property "for all $p \in U$ there exists an open neighbourhood $V \subseteq U$ of p and $s \in \mathcal{F}(V)$ such that for all $q \in V \quad f_q = s_q$" is the Sheafification of $\mathcal{F}$.

We will prove this in steps in the next lemmas.

Lemma 1.5.2. *$\mathcal{F}^{\#}$ is a Sheaf (of sets) and there exists a canonical Morphism of Presheaves $\mathcal{F} \to \mathcal{F}^{\#}$.*

Proof. Note that given $u \in U$ the condition $(*)$ depends only on the values of s_q for q in an open neighbourhood of p the map $\prod_{p\in U} \mathcal{F}_p \to \prod_{q\in V} \mathcal{F}_q$ sends elements of $\mathcal{F}^{\#}(U)$ to $\mathcal{F}^{\#}(V)$. Therefore taking this maps as restriction maps we make $\mathcal{F}^{\#}$ into a Presheaf.

Furthermore using the map described in Proposition 1.4.1 for every $V \subseteq U$ open we get the following commutative diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}^{\#}(U) & \rightarrowtail & \prod\limits_{p\in U} \mathcal{F}_p \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}(V) & \longrightarrow & \mathcal{F}^{\#}(V) & \rightarrowtail & \prod\limits_{q\in V} \mathcal{F}_q \end{array}$$

where the vertical arrows come from the restriction maps, giving a Morphism of Presheaves $\mathcal{F} \to \mathcal{F}^{\#}$.

To see that $\mathcal{F}^{\#}$ is indeed a Sheaf, consider an open set $U \subseteq X$ and an open covering $\{U_i\}_{i\in\mathcal{I}}$ of U.

For every $i \in \mathcal{I}$ let $s_i = (s_{i,p})_{p\in U_i} \in \mathcal{F}^{\#}(U_i)$ such that for all $i, j \in \mathcal{I} \quad s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$. Since $\prod_{p\in U} \mathcal{F}_p$ is a Sheaf we can get an element $s = (s_p)_{p\in U} \in \prod_{p\in U} \mathcal{F}_p$ by restricting s_i to U_i. Consider $p \in U$. Since $p \in U_i$ for some $i \in \mathcal{I}$ by $(*)$ there exists an open neighbourhood V of p and $t \in \mathcal{F}(V)$ such that for all $q \in V \quad s_{i,q} = (t, V)$. Therefore since $s_{i,q} = s_q$ we get that s verifies $(*)$. □

Lemma 1.5.3. *Let X be a topological space and let $\mathcal{F}$ be a Presheaf of sets over X. Then for every $p \in X$ we have $\mathcal{F}_p \cong \mathcal{F}^{\#}_p$.*

Proof. Let S be an arbitrary set. Then using the adjunctions Remark 1.5.2 and Proposition 1.5.1 we have

$$\begin{aligned} \mathrm{Hom}_{\mathsf{Set}}((\mathcal{F}^{\#})_p, S) &\cong \mathrm{Hom}_{\mathsf{Sh}}(\mathcal{F}^{\#}, \iota_{p,*}\underline{S}) \\ &\cong \mathrm{Hom}_{\mathsf{PSh}}(\mathcal{F}, \iota_{p,*}\underline{S}) \\ &\cong \mathrm{Hom}_{\mathsf{Set}}(\mathcal{F}_p, S) \end{aligned}$$

Therefore since this holds for every set S we have that the functors $\mathrm{Hom}_{\mathsf{Set}}((\mathcal{F}^{\#})_p, -)$ and $\mathrm{Hom}_{\mathsf{Set}}(\mathcal{F}_p, -)$ are isomorphic. Finally applying the Yonneda lemma we get $(\mathcal{F}^{\#})_p \cong \mathcal{F}_p$. □

Lemma 1.5.4. *Let $\mathscr{G}$ be a Sheaf of sets and let $\mathcal{F} \to \mathscr{G}$ be a Morphism of Presheaves. Then $\mathcal{F} \to \mathscr{G}$ factors uniquely as $\mathcal{F} \to \mathcal{F}^{\#} \to \mathscr{G}$.*

Proof. Consider the following commutative diagram

$$\begin{array}{ccccc} \mathcal{F} & \longrightarrow & \mathcal{F}^{\#} & \longrightarrow & \prod(\mathcal{F}) \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{G} & \longrightarrow & \mathcal{G}^{\#} & \longrightarrow & \prod(\mathcal{G}) \end{array}$$

We only need to prove that $\mathscr{G} \cong \mathscr{G}^{\#}$. By Stalks and isomorphisms we only need to show that for every $p \in X$ the Induced Stalk morphism is an isomorphism. But this is precisely Lemma 1.5.3. □

Should we mention the **espace étalé**?

Chapter 2

Ringed spaces and $\mathscr{O}_X$-modules.

2.1 Ringed spaces

Definition 2.1.1 (Ringed space)**.** A **ringed space** is a tuple $(X, \mathscr{O}_X)$ where X is a topological space and $\mathscr{O}_X$ is a Sheaf of rings over X.

We call $\mathscr{O}_X$ the *structure sheaf* of the ringed space $(X, \mathscr{O}_X)$. Sections of $\mathscr{O}_X$ over an open subset $U \subseteq X$ are called *functions on* U and functions on X are called (global) functions. ⌟

Example 2.1.1. Let and X be a topological space. Then the $\mathbb{R}$-algebra of continuous functions $\mathcal{C}^0_X(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous.}\}$ with the obvious restriction maps forms a Sheaf of rings over X. Therefore $(X, \mathcal{C}^0_X)$ is a Ringed space. ◊

Example 2.1.2. If for every point $p \in X$ the Stalks $\mathscr{O}_{X,p}$ of the structure sheaf are *local rings*[1] then we call $(X, \mathscr{O}_X)$ a **locally ringed space**. ◊

Definition 2.1.2 (Morphism of Ringed spaces)**.** Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two Ringed spaces. A **morphism of ringed spaces** $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a pair $(f, f^\sharp)$ where $f : X \to Y$ is a continuous function (i.e: a morphism in Top) and $f^\sharp : \mathscr{O}_Y \to f_*\mathscr{O}_X$ is an map of sheaves of rings (i.e: a morphism in $\mathsf{Sh}_{\mathsf{Ring}}(X)$), where $f_*\mathscr{O}_X$ is the Pushforward Sheaf. ⌟

Maybe it is better to define $f^\sharp$ as an f-map. [Stacks, Tag 008K]

Example 2.1.3. Let $f : X \to Y$ be a continuous function. Consider the Ringed spaces of continuous functions $(X, \mathcal{C}^0_X)$, $(Y, \mathcal{C}^0_Y)$.

Then the Morphism of Ringed spaces induced by f can be defined for every $V \subseteq Y$ open as $f^\sharp(V) : \mathcal{C}^0_Y(V) \to \mathcal{C}^0_X(f^{-1}(V))$, $f^\sharp(V)(h) = h \circ f|_{f^{-1}V}$. ◊

2.2 $\mathscr{O}_X$ modules

Definition 2.2.1 ($\mathscr{O}_X$-module)**.** Let $(X, \mathscr{O}_X)$ be a Ringed space. A **Sheaf of $\mathscr{O}_X$ modules** (or simply an $\mathscr{O}_X$-module) is a Sheaf $\mathscr{F}$ such that for every open set $U \subseteq X$ $\mathscr{F}(U)$ is an $\mathscr{O}_X(U)$-module compatible with the restriction maps, i.e: for every $f \in \mathscr{O}_X(U)$ and $s \in \mathscr{F}(U)$: $(f \cdot s)|_V = f|_V \cdot s|_V$. ⌟

Example 2.2.1. ... ◊

Definition 2.2.2 (Morphism of $\mathscr{O}_X$-modules)**.** Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. A morphism $\phi : \mathscr{F} \to \mathscr{G}$ is a Morphism of Sheafs that commutes with the multiplication of the structure sheaf $\mathscr{O}_X$. ⌟

[1] A commutative ring with unity R is a **local ring** if $0 \neq 1$ and for every a, b such that $a + b = 1$ either a or b is invertible.

Example 2.2.2. ... ◊

Therefore for every Ringed space $(X, \mathscr{O}_X)$ the collection of $\mathscr{O}_X$-modules forms a category $\mathscr{O}_X$-Mod.

2.2.1 $\mathscr{O}_X$-Mod and sheaves of abelian groups.

Just as abelian groups are $\mathbb{Z}$-modules, Sheafs of abelian groups can be seen as $\underline{\mathbb{Z}}_X$-modules where $\underline{\mathbb{Z}}_X$ is the Constant sheaf with coefficients in the integers.

To show this, for every Sheaf of abelian groups $\mathscr{F}$ over X and every open set $U \subseteq X$ we have to define an action $\cdot : \underline{\mathbb{Z}}_X(U) \times \mathscr{F}(U) \to \mathscr{F}(U)$ compatible with the restriction maps.

Let $f \in \underline{\mathbb{Z}}_X(U)$ and $s \in \mathscr{F}(U)$. Since f is locally constant there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U such that for every $i \in \mathcal{I}$ $\quad f|_{U_i} = n_i \in \mathbb{Z}$.

For every U_i consider $t_i = n_i \cdot s|_{U_i}$, where $n_i \cdot s|_{U_i}$ is shorthand for $s|_{U_i}$ added to itself n_i times. Note that this is well-defined because $\mathscr{F}(U_i)$ is an abelian group.

Let $U_i, U_j \in \{U_i\}_{i\in\mathcal{I}}$ such that $U_i \cap U_j = \varnothing$. For every $x \in U_i \cap U_j$ there exists an open neighbourhood $V \subseteq U_i \cap U_j$ of x were f is constant.

Therefore $n_i = n_j$ and $t_i|_{U_i\cap U_j} = t_j|_{U_i\cap U_j}$ and, applying axiom SH.2, we can define $f \cdot s = t$.

Checking the axioms in small enough open sets such that f is constant verifies that this action makes $\mathscr{F}$ an $\underline{\mathbb{Z}}_X$-module.

Do we have to actually write the check?

2.3 $\mathscr{O}_X$-Mod is abelian.

Definition 2.3.1 (Preadditive category)**.** A (small) category C is **preadditive** if every hom-set $\mathrm{Hom}(\triangle, \square)$ admits the structure of an abelian group and composition of morphisms is bilinear with respect to the group operation. ⌟

Definition 2.3.2 (Additive category)**.** A category C is **additive** if it is a Preadditive category, has a zero object and has all binary biproducts. ⌟

Definition 2.3.3 (Abelian category)**.** A Additive category C is **abelian** if

- It has all kernels and cokernels.
- Every monomorphism is the kernel of some morphism and every epimorphism is the cokernel of some morphism.

⌟

Move this definitions to the Homological Algebra section

Let's check first that $\mathscr{O}_X$-Mod is a Preadditive category.

Lemma 2.3.1. *The category $\mathscr{O}_X$-Mod is a Preadditive category.*

Proof. Let $\phi : \mathscr{F} \to \mathscr{G}$ and $\psi : \mathscr{F} \to \mathscr{G}$ be two Morphisms of Ringed spaces.

For each open subset $U \subseteq X$ define $(\phi + \psi)(U) = \phi(U) + \psi(U)$. □

Now Additive category

Lemma 2.3.2. *The category $\mathscr{O}_X$-Mod admits a zero object.*

Proof. Consider the $\mathscr{O}_X$ module that sends every open set to $\{0\}$. □

Lemma 2.3.3. *The category $\mathscr{O}_X$-Mod has all binary biproducts. We define their direct sum a $\mathscr{F} \oplus \mathscr{G} = \mathscr{F} \times \mathscr{G}$.*

Proof. Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. □

Theorem 2.3.1. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the kernel of φ is the subsheaf of $\mathscr{F}$ with sections*

$$(\ker \varphi)(U) = \{s \in \mathscr{F}(U) \mid \varphi(s) = 0\}$$

Furthermore for every $x \in X$ $\quad (\ker \varphi)_x = \ker \varphi_x$.

Proof. If $V \subseteq U$ □

$$\begin{array}{ccccccc} 0 & \longrightarrow & (\ker\varphi)(U) & \longrightarrow & \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ & & \vdots & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ 0 & \longrightarrow & (\ker\varphi)(V) & \longrightarrow & \mathscr{F}(V) & \xrightarrow{\varphi(V)} & \mathscr{G}(V) \end{array}$$

Theorem 2.3.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the cokernel of φ is the Sheafification of the quotient Presheaf*

$$(\mathrm{coker}_{\mathrm{pre}}\, \varphi)(U) = {}^{\mathscr{G}(U)}\!/_{\varphi(\mathscr{F}(U))}$$

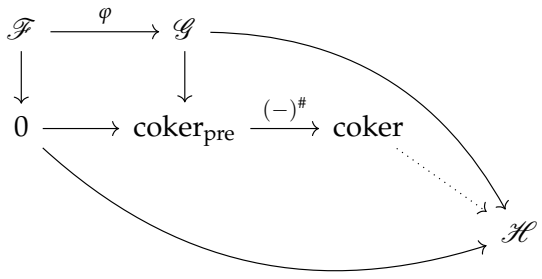
Prove that this is the cokernel in PSh.

Furthermore for every $x \in X$ $\quad (\mathrm{coker}\, \varphi)_x = \mathrm{coker}\, \varphi_x$.

Proof. Given a commutative diagram

$$\begin{array}{ccc} \mathscr{F} & \xrightarrow{\varphi} & \mathscr{G} \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathscr{H} \end{array}$$

We construct □



Therefore we have proven that $\mathscr{O}_X$-Mod is an Abelian category and in particular by $\mathscr{O}_X$-Mod and sheaves of abelian groups $\mathsf{Sh}_{\mathsf{Ab}}(X)$ is an Abelian category.

2.4 Sequences of $\mathscr{O}_X$-Mod

Theorem 2.4.1. *The sequence of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for every $x \in X$ the sequence $\mathscr{F}_x \to \mathscr{G}_x \to \mathscr{H}_x$ is exact at $\mathscr{G}_x$.*

Probably not a *theorem* but a direct consequence of the commutatitivity of stalks and (co)kernels

Maybe talk about resolutions / types of resolutions here?

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